

## 1.4 Materials

This topic covers flow of liquids, viscosity, Stokes' law, properties of materials, Hooke's law, Young's modulus and elastic strain energy.

This topic may be taught using, for example, a case study of the production of sweets and biscuits. It could also be taught using the physics associated with spare part surgery for joint replacements and lens implants.

Learning outcomes 18–27 should be studied using variety of applications, for example, making and testing food, engineering materials, spare part surgery. This unit includes many opportunities to develop experimental skills and techniques.

| Students will be assessed on their ability to:                                                                                                                                                                                                    | Suggested experiments                                     |
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| 18 understand and use the terms <i>density</i> , <i>laminar flow</i> , <i>streamline flow</i> , <i>terminal velocity</i> , <i>turbulent flow</i> , <i>upthrust</i> and <i>viscous drag</i> , for example, in transport design or in manufacturing |                                                           |
| 19 recall, and use primary or secondary data to show that the rate of flow of a fluid is related to its viscosity                                                                                                                                 |                                                           |
| 20 recognise and use the expression for Stokes's Law, $F = 6\pi\eta rv$ and upthrust = weight of fluid displaced                                                                                                                                  |                                                           |
| 21 investigate, using primary or secondary data, and recall that the viscosities of most fluids change with temperature. Explain the importance of this for industrial applications                                                               |                                                           |
| 22 obtain and draw force–extension, force–compression, and tensile/compressive stress-strain graphs. Identify the <i>limit of proportionality</i> , <i>elastic limit</i> and <i>yield point</i>                                                   | Obtain graphs for, example, copper wire, nylon and rubber |

| Students will be assessed on their ability to:                                                                                                                                                                                                                                                               | Suggested experiments                                        |
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| 23 investigate and use Hooke's law, $F = k\Delta x$ , and know that it applies only to some materials                                                                                                                                                                                                        |                                                              |
| 24 explain the meaning and use of, and calculate <i>tensile/compressive stress, tensile/compressive strain, strength, breaking stress, stiffness</i> and <i>Young Modulus</i> . Obtain the Young modulus for a material                                                                                      | Investigations could include, for example, copper and rubber |
| 25 investigate elastic and plastic deformation of a material and distinguish between them                                                                                                                                                                                                                    |                                                              |
| 26 explore and explain what is meant by the terms <i>brittle, ductile, hard, malleable, stiff</i> and <i>tough</i> . Use these terms, give examples of materials exhibiting such properties and explain how these properties are used in a variety of applications, for example, safety clothing, foodstuffs |                                                              |
| 27 calculate the elastic strain energy $E_{el}$ in a deformed material sample, using the expression $E_{el} = \frac{1}{2} Fx$ , and from the area under its force/extension graph                                                                                                                            |                                                              |